

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458544.8367 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.044
Transit depth (Rp/Rs)^2	1.54 +/- 1.1 [%]
Orbital Inclination (inc)	88.43 +/- 1.89
Airmass coefficient 1 (a1)	1.003 +/- 0.038
Airmass coefficient 2 (a2)	0.0025 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	3.74 %
Best Comparison Star	#2 - [186, 85]
Optimal Aperture	3.59
Optimal Annulus	8.3
Transit Duration (day)	0.09 +/- 0.054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.044 vs 0.1295 (deviation 0.13σ)
Duration fit vs published	0.09 d vs 0.1146 d (deviation 0.46σ)
Mid-time O–C	5.3 min
Depth SNR	2.8
Residual scatter	3.74 %

Light curve

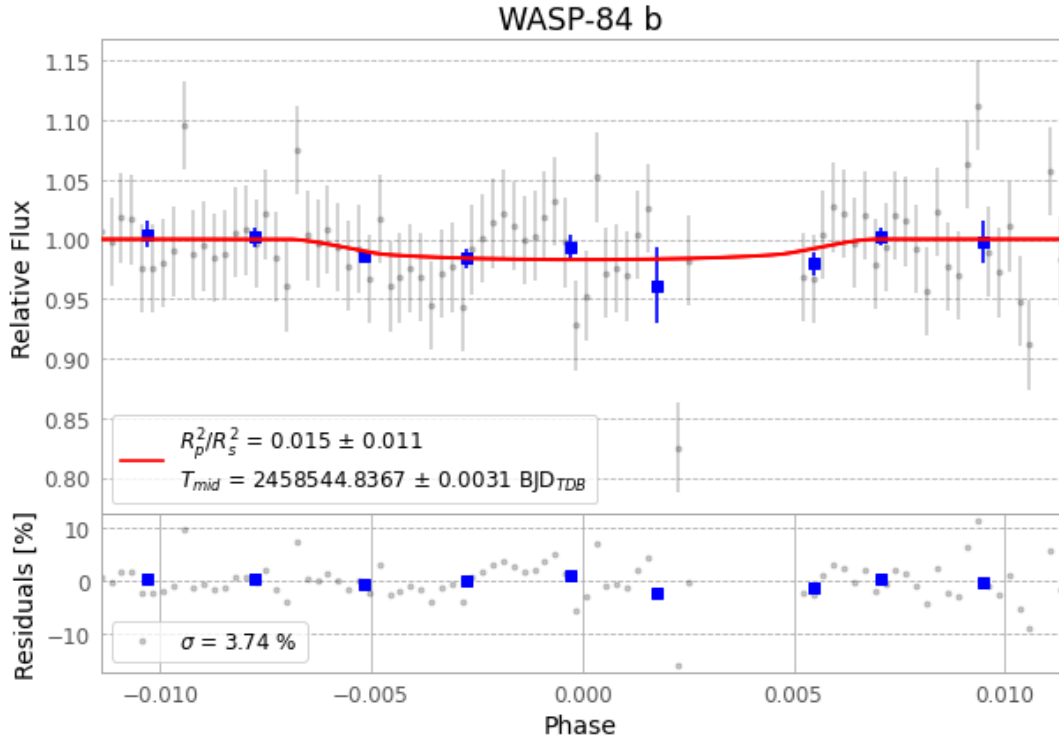


Figure 1 — FinalLightCurve_WASP-84 b_2019-03-01.png (SHA-256 6b44a8ec41d10452...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5dff36d4570ced1a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

95 FITS frames (62 MB), WASP-84190302053612.FITS → WASP-84190302101814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-190302.log (2bfb36149478...), FinalLightCurve_WASP-84 b_2019-03-01.png (6b44a8ec41d1...), FinalLightCurve_WASP-84 b_2019-03-01.csv (2c66992c4f90...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:29Z.